Arrays:

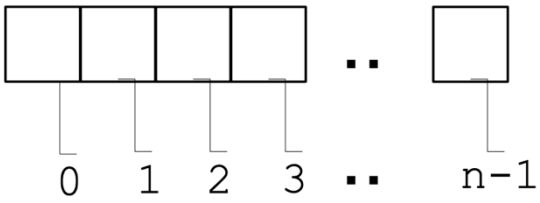
An array is an indexed list of values that all have the **same type**.

You can make an array of any type int, double, String, etc.

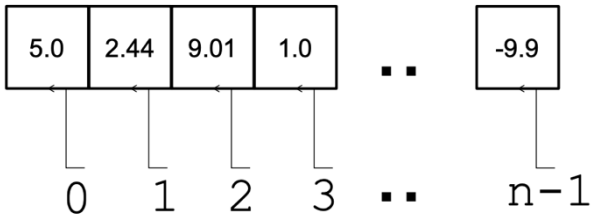
In array, the memory allocated is contiguous.

1. One-dimensional Array:

It is an array that has one index. **The index starts at zero and ends at length-1.**



Following an example of array of type double:



How to declare, initialize and access array?

An array is **defined** as following:

data_type array_name[];

OR

data_type[] array_name;

```
int[] values; // declare array of type int named “values”
int values[]; // same as above
```

To create an array of a given size, here size specifies the number of elements in the array, use the operator **new**:

```
double[] c = new double[5]; // declare an double array named “c” of size 5
double c[] = new double[5]; // same as above
```

Curly braces { } can be used to **initialize** an array. It can **ONLY** be used when you declare the variable.

```
int[] values = { 12, 24, -23, 47 }; // size of array deduced from the number of items
int values[] = { 12, 24, -23, 47 }; // same as above
```

Each array has a **length** variable built-in that contains the length of the array.

```
int[] values1 = new int[12]; // Declare and allocate a 12 elements int
array
int num = values1.length; // num is 12

int[] values2 = { 1, 2, 3, 4, 5 }
int size = values2.length; // size is 5
```

To access the elements of an array, use the [ ] operator. This operator contains the index value.

```
int[] values = { 12, 24, -23, 47 };
values[3] = 18; // {12, 24, -23, 18}
int x = values[1] + 3; // x = 27
```

Note: If the index is either outside the range of [0, length-1] or negative, Java Runtime will signal an exception called **ArrayIndexOutOfBoundException**.

Q> Is there an error in the following codes?

```
1. int[] values = {1, 2.5, 3, 3.5, 4};
2. char[] array = {'a', '*', '4', '/'};
   System.out.println(array[4]);
```

You can process all the elements of an array via a **loop.**

```
int[] values = { 12, 24, -23, 47 };
for (int i=0; i < values.length; i++)
{
System.out.println(values[i]);
}
```

```
12
24
-23
47
```

```
int[] values = { 12, 24, -23, 47 };
int i = 0;
while (i < values.length)
{
System.out.println(values[i]);
i++;
}
```

```
12
24
-23
47
```

```java
/* Java program to illustrate creating an array of type double and
   print each element with its index */
public class ArrayExample1 {
     public static void main(String[] args) {
     double[] array= {22.4, 13.3, -6.5, 9};
     for (int c = 0; c < array.length; c++)
     System.out.println("Element at index "+ c + " = " + array[c]);
     }
}
```

```
Problems | Javadoc | Declaration | Console | Progress
<terminated> ArrayExample1 [Java Application] /Library/Java/JavaVirtualMachin
Element at index 0 = 22.4
Element at index 1 = 13.3
Element at index 2 = -6.5
Element at index 3 = 9.0
```

Example: Write a Java program to find the summation of the elements in an array then print out the square root of that summation.

```java
public class ArrayExample2 {
     public static void main(String[] args) {
     int numbers[] = {8, 2, 6, 4, 3};
     int sum = 0;
     for (int i=0; i < numbers.length; i++)
          sum = sum + numbers[i];
     System.out.println("The summation is: " + sum);
     System.out.println("The square root is: " + Math.sqrt(sum));
     }
 }
```

```
Problems | Javadoc | Declaration | Console | Progress
<terminated> ArrayExample2 [Java Application] /Library/Java/JavaVirtualMachin
The summation is: 23
The square root is: 4.795831523312719
```

```java
// Java Example Read and Print Array
import java.util.Scanner;
public class ArrayExample3 {
   public static void main(String[] args) {

    // Declare variables
      int size;
      int[] items;
      Scanner in = new Scanner(System.in);

    // Enter the number of items i.e. the size of the array
      System.out.print("Enter the number of items: ");
      size = in.nextInt();

      items = new int[size]; // Allocate the array

      // Read the items of the array
      System.out.print("Enter the value of all items (separated
by space): ");
      for (int i = 0; i < items.length; i++)
         items[i] = in.nextInt();

      in.close();

    // Print array contents
      System.out.print("The array contains: ");
      for (int i = 0; i < items.length; i++)
       System.out.print(items[i] +" ");
   }
}
```

Problems | Javadoc | Declaration | Console | Progress

```
<terminated> ArrayExample3 [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_112.jdk/Contents
Enter the number of items: 5
Enter the value of all items (separated by space): 33 8 25 -4 7
The array contains: 33 8 25 -4 7
```

Example 1: Write a Java program to store and display the maximum daily temperatures in a week.

```java
import java.util.Scanner;
public class TemperatureReadings {
     public static void main(String[] args) {
     Scanner keyboard = new Scanner(System.in);
     double[] temperature; // Create the array
     temperature = new double[7]; // Allocate a 7 elements
     // Enter temperatures
     for (int i = 0; i < temperature.length; i++)
     {
     System.out.println("enter max temperature for day " + (i+1));
     temperature[i] = keyboard.nextDouble();
     }
     keyboard.close();
     // Display Temperatures
     System.out.println(); // blank line
     System.out.println("***TEMPERATURES ENTERED***");
     System.out.println(); // blank line
     for (int i = 0; i < temperature.length; i++)
          System.out.println("day "+(i+1)+" "+ temperature[i]);
     }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> TemperatureReadings [Java Application] /Library/Java/JavaVirtualM
enter max temperature for day 1
23
enter max temperature for day 2
25
enter max temperature for day 3
22
enter max temperature for day 4
26
enter max temperature for day 5
27
enter max temperature for day 6
25
enter max temperature for day 7
30

***TEMPERATURES ENTERED***

day 1 23.0
day 2 25.0
day 3 22.0
day 4 26.0
day 5 27.0
day 6 25.0
day 7 30.0
```

Example 2: Write a Java program to store a week temperature and print the highest temperature.

```java
import java.util.Scanner;
public class HighestTemperature {
	public static void main(String[] args) {
	Scanner keyboard = new Scanner(System.in);
	double[] temperature; // Create the array
	temperature = new double[7]; // Allocate a 7 items in the array
	// Enter Temperatures
	for (int i = 0; i < temperature.length; i++)
	{
	System.out.println("enter max temperature for day " + (i+1));
	temperature[i] = keyboard.nextDouble();
	}
	keyboard.close();

	double highest=temperature[0];// Assume the first item is the highest
	//This loop runs from the 2nd item to the last item in the array
	for (int i = 1; i < temperature.length; i++)
		if (temperature[i] > highest)
			highest=temperature[i]; // Reset result to new highest
	System.out.println("The highest temperature of the week " + highest);
	}
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> HighestTemperature [Java Application] /Library/Java/JavaVirtualM
enter max temperature for day 1
25
enter max temperature for day 2
27
enter max temperature for day 3
26
enter max temperature for day 4
30
enter max temperature for day 5
29
enter max temperature for day 6
28
enter max temperature for day 7
24
The highest temperature of the week 30.0
```

Example 3: Write a Java program to print even numbers in an array.

```
public class ArrayEvenNumber {
	public static void main(String[] args) {
	int numbers[] = {5, 2, 6, 3, 4};
	for (int i=0; i < numbers.length; i++)
		if (numbers[i] %2 == 0)
			System.out.println(numbers[i]);
	}
}
```

```
Problems  @ Javadoc  Declaration  Console  Progress
<terminated> ArrayEvenNumber [Java Application] /Library/Java/JavaVirtualMac
2
6
4
```

Example 4: Write a Java program to print the average of numbers in an array.

```
public class ArrayAverage {
	public static void main(String[] args) {
	double[] array= {22.4, 13.3, 9.5, 6, 10.3};
	double sum=0;
	for (int c = 0; c < array.length; c++)
		sum=sum+array[c];
	System.out.println("The average = " + sum/array.length);
	}
}
```

```
Problems  @ Javadoc  Declaration  Console  Progress
<terminated> ArrayAverage [Java Application] /Library/Java/JavaVirtualMachines
The average = 12.3
```

Example 5: Write a Java program to search an element in an array.

```
import java.util.Scanner;
public class ArraySearch {
	public static void main(String[] args) {
	int numbers[] = {5, 2, 7, 4, 10};
	int i,num,flag=0;
	Scanner s = new Scanner(System.in);
	System.out.print("Enter the element you search for : ");
	num = s.nextInt();
	s.close();

	for (i=0; i < numbers.length; i++)
		if (num==numbers[i])
		{
			flag=1;
			break;
		}
```

```
        if (flag==1)
            System.out.println("The Element Found at position: "+ i);
        else
            System.out.println("The Element Not Found");
        }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 7
The Element Found at position: 2
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 3
The Element Not Found
```

Example 6: Write a Java program to sort the items of an array in ascending order.

```
public class ArrayAscendingOrder {
        public static void main(String[] args) {
            double[] array= {22.4, 13.3, 9.5, 6, 10.3};
            double temp;

          for (int i = 0; i < array.length; i++)
          {
              for (int j = i + 1; j < array.length; j++)
              {
                  if (array[i] > array[j])
                  {
                      temp = array[i];
                      array[i] = array[j];
                      array[j] = temp;
                  }
              }
          }
          System.out.print("Ascending Order: ");
          for (int i = 0; i < array.length; i++)
            System.out.print(array[i] + " ");
      }
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> ArrayAscendingOrder [Java Application] /Library/Java/JavaVirtualM
Ascending Order: 6.0 9.5 10.3 13.3 22.4
```

Homework:

Q> Write a Java program to count the number of occurrences of an element in an array.

Q> Write a Java program to print the prime numbers in an array.

Q> Write a Java program to sort the items of an array in descending order.

Q> Write a Java program to sort names of an array in an alphabetical order.

Lecture //Programs Examples on 1 D Array

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Q1: Write Java program to create array have 5 string items {Ali, Hassan , Mohaamed, Jaafar}

Then search for word Ali , if his name in array , **print found if not print not found**.

```
public class find {
   /**
    * @param args the command line arguments
    */
   public static void main(String[] args) {
      // TODO code application logic here
       String names[]={"Ali","Hassah","Jaafar","Mohamed"};
      int x=0;// flag
      for(int i=0;i<names.length;++i)
      {    if(names[i]=="Ali")
           x=1;        }
      if (x==1) System.out.println("Found");
      else System.out.println("not Found");
   }   }
```

Q2: Write java codes to create array with any five integer items { 5,4,2,1} then sorting array in ascending order.

```
public class SortArr{
**/
param args the command line arguments@ *
```

```
/*
   public static void main(String[] args){
 //        TODO code application logic here
        int []a;{5,4,2,1}=

        int swap;
        for(int i=0;i<a.length;++i)
        {
        for(int j=i+1;j<a.length;++j)
            if (a[i]>a[j])
            {
                swap=a[i];
                a[i]=a[j];
                a[j]=swap ;
{
{
 //        for print
        for(int i=0;i<a.length;++i)
        System.out.println(a[i]);

}


}
```

Q3: Write java codes to create array with any five integer items {10,20,60,70,30} then find Max and Min Numbers in array

H.W//

1- Rewrite all above programs and input items from Keyboards

2- Rewrite second program and input char instead of string from keyboard

End

H.W 1D Array 2026

1. Create integer array (`1, 2, 2, 3, 3, 3)` **Count Frequency of Each Element:**

---

2. Create integer array (`0, 1, 0, 3, 12) then` **Shift All Zeroes to the End:**

---

3. Create any two integer array from Keyboard , then Merge Two Sorted Arrays

---

4. Create array (`1, 2, 3, 4, 5, 6) then` **Split an Array into Two Halves:**

---

5. Create array from Key board then Sum of All Elements:

---

6. Create array (`10, 20, 4, 45, 99) then ,` **Find Second Largest Element Without Sorting:**

---

7. Create any two arrays from keyboard then Check if Two Arrays Are Equal:

---

8. Create array then rotate Array to Left by Given Positions =2

---

9. Create array from the keyboard then print the array from second position after that Find Missing number in Consecutive Integers:

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10. create array from keyboard then Calculate Average Value